SIW Bandpass Filter — Project State

2.87 GHz dual-mode SIW filter on RO4350B

2026-09-02

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The goal

A Substrate Integrated Waveguide (SIW) bandpass filter:

Parameter	Target
Centre frequency	2.87 GHz
Bandwidth	143.5 MHz (5 % FBW)
Return loss	at most −16 dB in band
Response	Chebyshev, 0.1 dB ripple
Substrate	Rogers RO4350B, 0.76 mm, 0.5 oz copper (17.5 µm)
Vias	Ø1.5 mm, 3.0 mm pitch
Connector	Taoglas EMPCB.SMAFSTJ.B.HT, CPWG launch

For what to do next, see `PLAN.md`. Superseded documents are in `docs/archive/`.

The design we are building

Dual-mode N=2 — a single square SIW cavity carrying two degenerate modes, split by an internal floating iris row.

Geometry: `a = 57.21`, `w_iris = 15.26`, `x_iris = 28.68`, `y_wall = 8.62`, `w_ext = 16.31` mm.

Why this one, and not a cascade

Three candidates were compared head to head on identical axes:

	dual-mode N=2	cascade N=3	cascade N=4
Return loss	**−17.7 dB**	−14.9 dB	−6.1 dB
Insertion loss	**−1.42 dB**	−1.85 dB	−2.7 dB
Board length	**119 mm**	172 mm	211 mm
Spurious to 8 GHz	**clean**	2nd passband nearly as strong as the main one	same problem

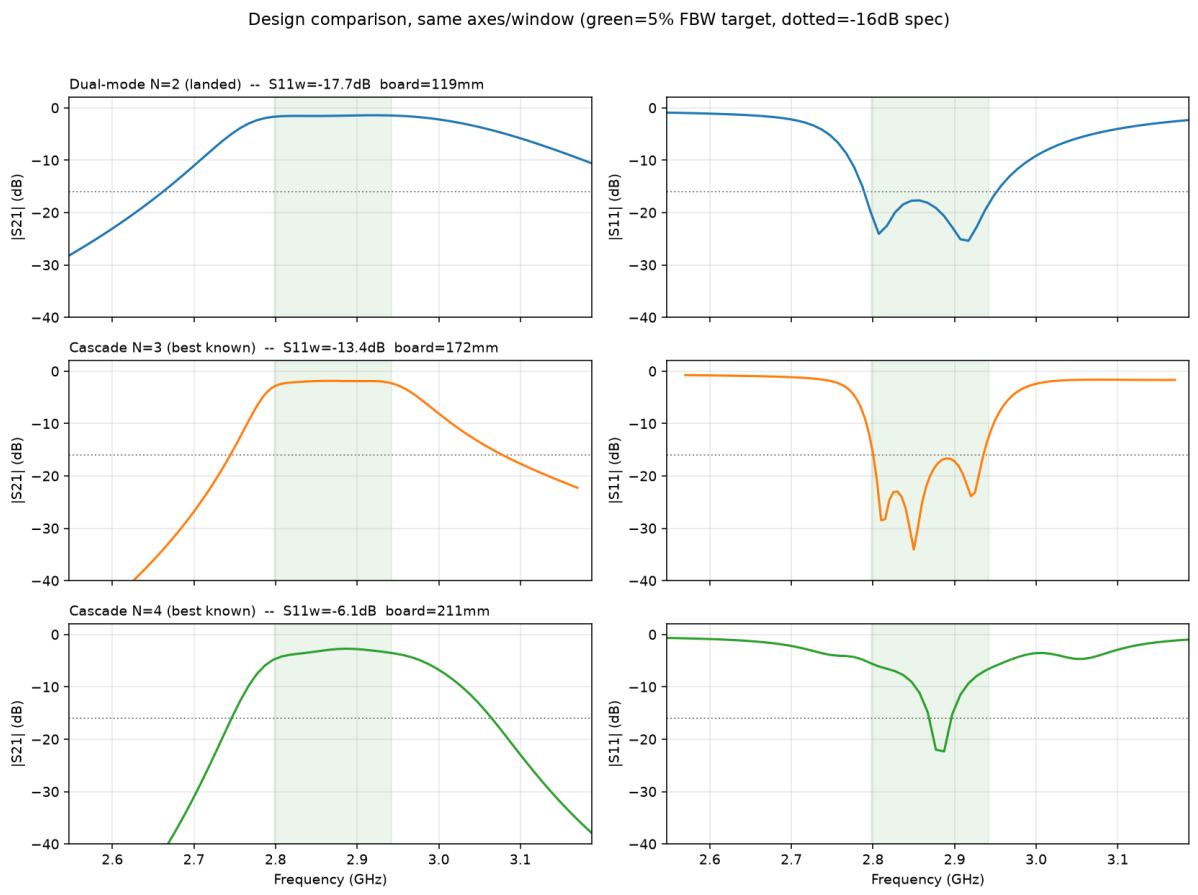


Figure 1: Dual-mode N=2 versus the two cascade designs, same axes. The dual-mode response has the deepest return-loss nulls; the cascades were rejected mainly for their spurious passbands, visible in the broadband check.

The cascade designs are **parked, not abandoned** — they are a real comparative result, and their spurious passbands are the reason they were not chosen.

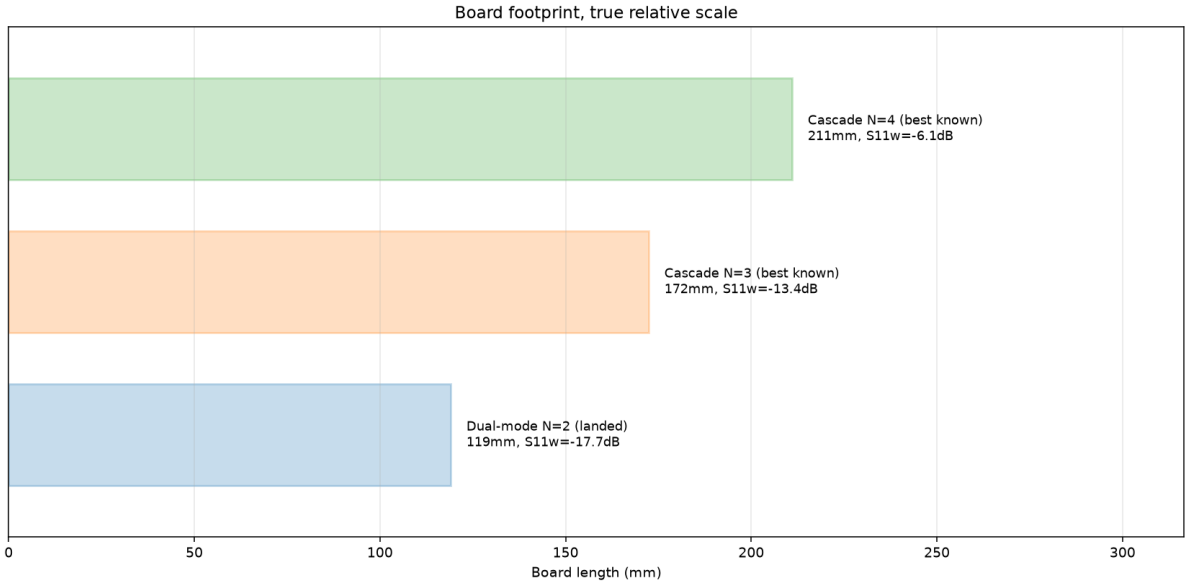


Figure 2: Board footprint at true relative scale. The dual-mode design is roughly half the length of the N=4 cascade.

The fabrication stack

The board is modelled with its real manufacturing layers, not as idealised metal: lossy copper, ENIG finish, soldermask, silkscreen, and filled or open via barrels are all genuine geometry and materials.

What the stack costs

Variant	fc (GHz)	IL (dB)	BW (MHz)	Return loss (dB)
PEC (idealised)	2.911	−1.42	318	−17.69
Real stack, vias tented	2.909	−1.79	298	−12.48
Real stack, vias open	2.909	−1.79	298	−12.47
Real stack, no silkscreen	2.907	−1.79	297	−12.50

Three findings, each measured:

1. **Return loss is the fabrication penalty.** The dielectric stack over the feed detunes the launch by roughly 5 dB. Insertion loss is a side issue.
2. **Real copper costs about 0.37 dB of insertion loss** that a perfect-conductor model omits entirely.
3. **Via tenting and silkscreen are electrically irrelevant** — all three real-stack variants agree to within 0.03 dB, below solver noise. Do not pay for via filling on RF grounds.

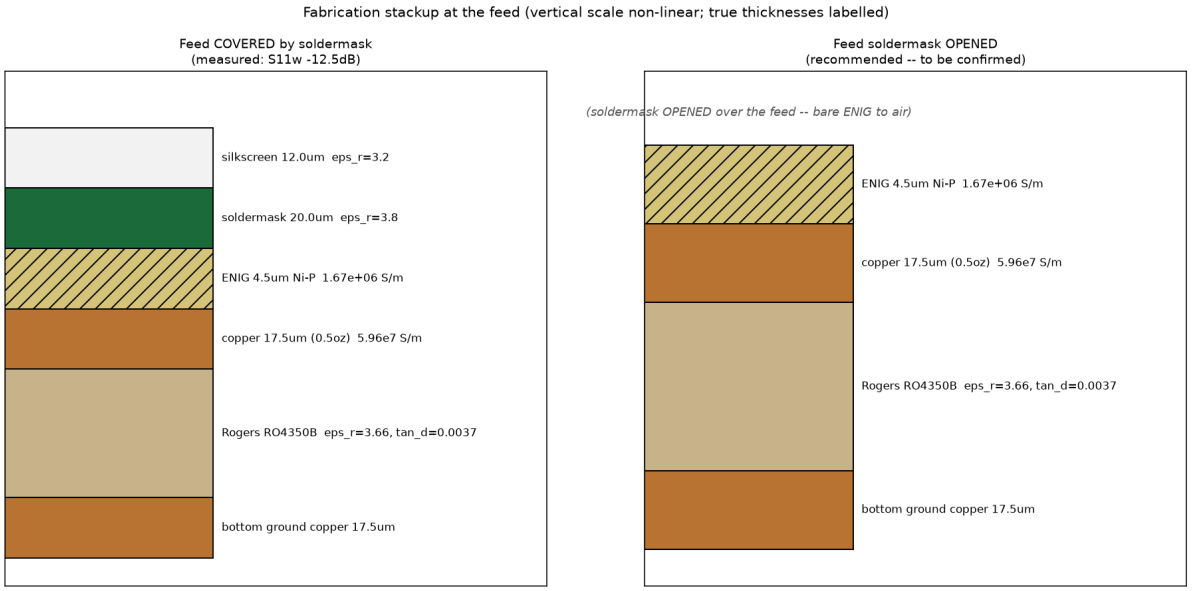


Figure 3: Fabrication stack at the feed. Vertical scale is non-linear so the thin finish layers remain visible; true thicknesses are labelled. Left: feed covered by soldermask. Right: soldermask opened over the feed, the recommended option.

What is true right now

Solid

- The dual-mode geometry meets the return-loss target **under an idealised model** (PEC metal, pad-only launch): −17.7 dB, 2 reflection zeros, clean to 8 GHz.
- The fabrication stack is modelled for real, and its cost is quantified above.

Not true yet

- **Bandwidth is about 2× the specification** (about 300 MHz against 143.5 MHz). This is the oldest open problem, and no fabrication choice affects it. It requires the cavity coupling to be re-derived, not a launch adjustment.
- **Nothing has been optimised under the CPWG launch.** It was merged into the dual-mode model on 2026-09-02; every dual-mode measurement predating that used the pad-only launch and is archived in `golden/dualmode_N2/archive_padonly_launch/`.
- **No board has been ordered.** No CST export, no gerbers, no DRC check.

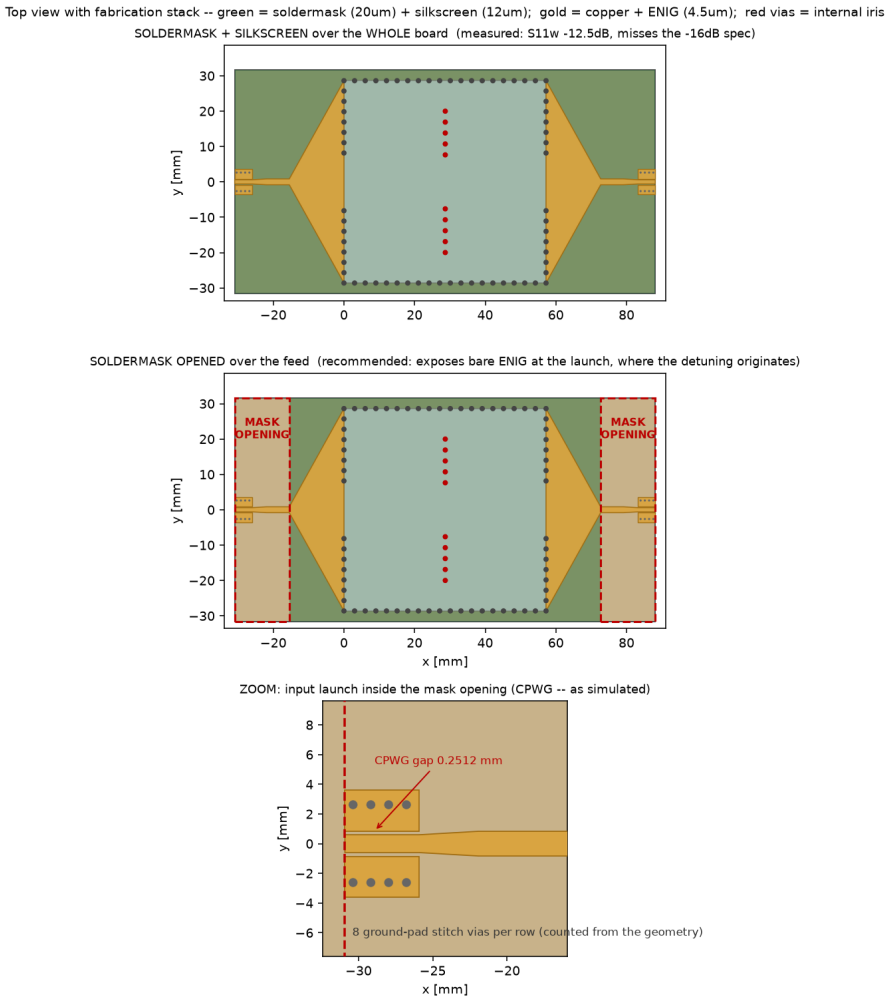


Figure 4: Top view showing where each layer sits, including the impedance-matched CPWG launch (0.2512 mm gap, 6 stitch vias per side) and the pad taper into the 50 Ω microstrip. The bottom panel zooms on the launch, where the coplanar gap is sub-pixel at board scale.

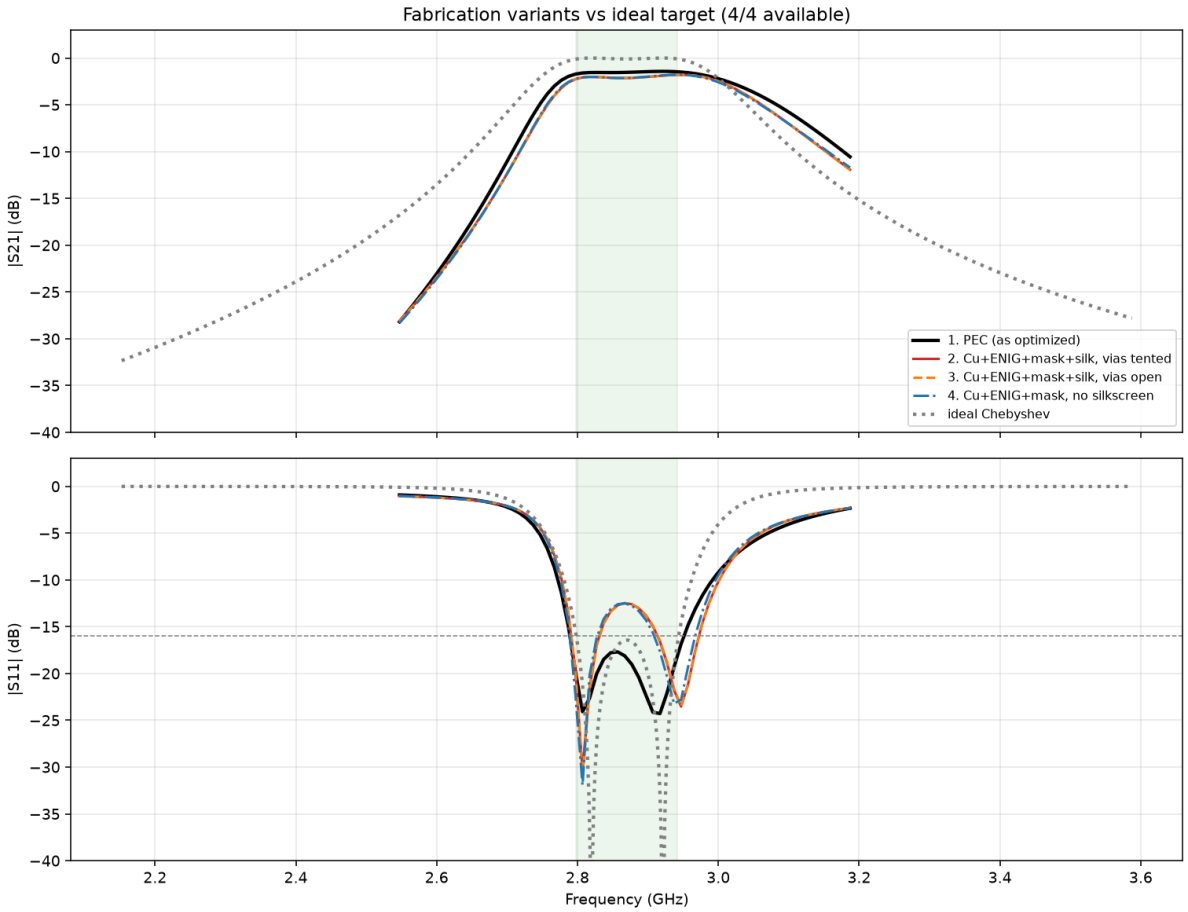


Figure 5: Idealised PEC metal versus the real fabrication stack, with the ideal Chebyshev target for reference. Note these runs used the earlier pad-only launch — the mechanisms hold, the absolute numbers are being re-measured.

Building this document

Pandoc-compatible with YAML front matter and relative image paths.

```
cd docs && uv run gen_pdfs.py --force STATE.md    # needs xelatex
```

`xelatex` is not installed on the development machine, so the PDF has to be built where a LaTeX toolchain exists. Any pandoc target works without it, for example a single self-contained HTML file with the figures embedded:

```
pandoc STATE.md -o state.html --standalone --embed-resources --toc
```

How to reproduce these figures

```
siw stackup --verify       # stack cross-section and top view, plus a check
                           # that the drawing matches the meshed geometry
siw compare-designs        # dual-mode versus the cascade designs
siw compare-fab-variants   # idealised metal versus the real stack
siw layout --topology dualmode --N 2 \
    --params a=57.21,w_iris=15.26,x_iris=28.68,y_wall=8.62,w_ext=16.31
```

The one habit that matters

Seven silent bugs were found in this project's own model. Not one raised an error; every one produced plausible numbers that were wrong:

- a layer was added to the mesh but never given a material, so its faces carried no boundary condition — the solver warned, computed anyway, and returned *less* loss than a lossless conductor;
- a figure was drawn from a different preset than the mesh used, so published pictures showed a launch that had never been simulated;
- an adapter silently dropped feature data, so two fabrication studies ran with the feature switched off.

Each was caught by checking a number against physics, never by the code failing. **Before trusting any result, check it against what physics says it should be.** Insertion loss cannot beat a lossless model. Pole count must match the filter order. A hole far below its cutoff frequency cannot radiate.

Guards now in the code:

- `_driven_config` raises if a mesh domain has no material;
- the mesh builder reports any unclassified exterior face;
- `siw stackup --verify` compares the drawing against the meshed geometry vertex by vertex.